

Task 4 | Retraction-associated FASTQ data support source-level audit, not claim validation

A

Retraction-associated public FASTQ case

Used as a negative-control audit,
not as evidence for the retracted claim.

Publication

Genes Immunity
PMID 39533019

Retraction note

Genes Immunity
PMID 41927935

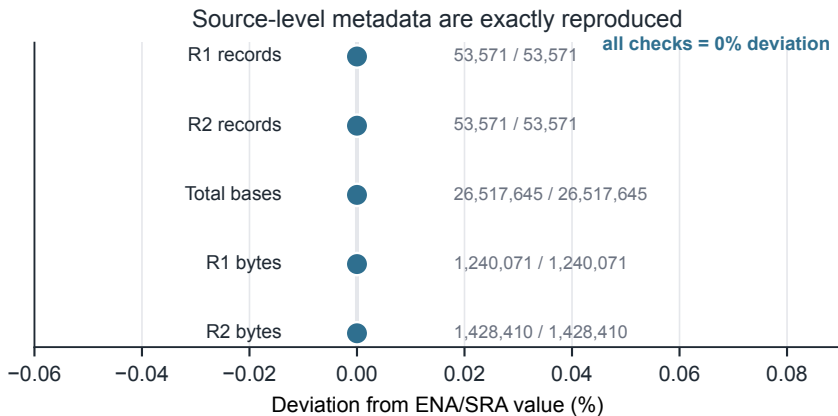
Public record

SRP508771
PRJNA1111407
SRR29088443

Sequencing file

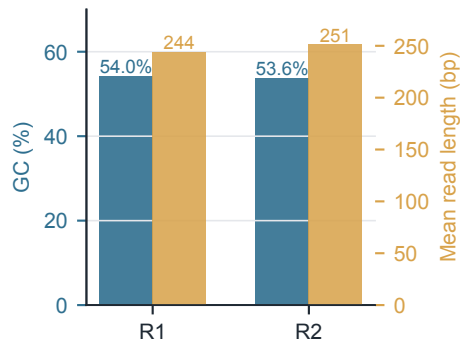
Paired FASTQ
Escherichia sp.

B



C

Derived FASTQ quality metrics



D

Audit interpretation

- FASTQ format
pass
- Paired-end structure
pass
- Read/base/file metadata
match public record
- QC report generation
FastQC + MultiQC + seqkit
- Biological conclusion
not claimed

Negative-control role:
audit source-data reproducibility
from a problematic literature record.